

Section A – (Start a new Booklet)

Question 1. (18 marks)

Marks

(a) If $u = 4 + 2i$ and $v = 3 - 5i$ find:

(4)

(i) $iu + \bar{v}$

(ii) $\operatorname{Re}\left(\frac{1}{v}\right)$

(b) (i) Express each of the complex numbers $z = 2i$ and $w = 1 + \sqrt{3}i$ in modulus – argument form.

(2)

(ii) Indicate the points representing the complex numbers $z = 2i$ and $w = 1 + \sqrt{3}i$ on an Argand diagram.

(2)

(iii) Find the exact values of $\arg\left(\frac{z}{w}\right)$ and $\arg(z + w)$

(4)

(c) The polynomial equation $x^3 - 3x^2 + x - 5 = 0$ has roots α, β and γ . Find the value of $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$

(3)

(d) Find $\int x \tan^{-1} x \, dx$

(3)

End of Question 1.

Question 2. (16 marks)

Marks

- (a) If α, β and γ are roots of $x^3 - 4x - 1 = 0$ find the polynomial equation with roots $2\alpha, 2\beta$ and 2γ (answer in expanded form). (3)

- (b) (i) Express $\sqrt{8-6i}$ in the form $a + ib$ where a and b are real and $a > 0$. (3)

- (ii) Hence solve the equation (3)

$$2z^2 + (1-3i)z - 2 = 0$$

expressing the answers in the form $X + iY$ where X and Y are real.

- (c) Let $A = 1 + 2i$ and $B = 2 - i$.

On an Argand diagram sketch the locus specified by:

- (i) $\arg(z + B) = \frac{3\pi}{4}$ (2)

- (ii) $|z - A| = |z - B|$ (2)

- (d) Find $\int \frac{x+5}{\sqrt{x+6}} dx$ (3)

End of Question 2.

END OF SECTION A.

Section B – (Start a new Booklet)

Question 3. (14 marks)

Marks

(a) Find:

(i) $\int \frac{dx}{\sqrt{3+2x-x^2}}$ (2)

(ii) $\int \frac{dx}{(x+1)(x^2+4)}$ (3)

(iii) $\int \frac{e^x + e^{2x}}{1 + e^{2x}} dx$ (3)

(b) (i) Given that (4)

$$u_n = \int_0^1 (1-x^2)^{\frac{n}{2}} dx$$

prove that $u_n = \frac{n}{n+1} u_{n-2}$

(ii) Evaluate u_5 (2)

End of Question 3.

Question 4. (15 marks)

Marks

- (a) If $z = 1 + i$ is a root of the equation

$$z^3 + pz^2 + qz + 6 = 0$$

where p and q are real, find:

- (i) the values of p and q (3)
- (ii) all the roots of the equation. (1)
- (b) Sketch the region in the Argand diagram consisting of those points representing the complex number z which simultaneously satisfy $|\arg z| < \frac{\pi}{3}$ and $z + \bar{z} < 4$ and $|z| > 2$ (4)
- (c) The complex number z represents a variable point P in the Argand plane. If $|z| = |z - \alpha|$ where α is real evaluate $\arg(z^2 - z\alpha)$ (3)
- (d) Given that w is the non-real root of $z^5 - 1 = 0$ with smallest positive argument find a real quadratic equation with roots w^2 and w^3 . (4)

End of Question 4.

END OF SECTION B.

Section C – (Start a new Booklet)

Question 5. (12 marks)

Marks

- (a) Four medical tests A, B, C and D are carried out within 14 days on a patient. A must precede B and B must precede C and D. On the days when A and B are carried out the patient must not undergo any other test, but C and D can be carried out on the same day or on different days in any order.

In how many ways can the days for the test be chosen? (4)

- (b) K is a positive real number and u, v are complex numbers. Show that the points on the Argand diagram which represent respectively the numbers:

$$u, v, \frac{u - iKv}{1 - iK}$$

form the vertices of a right-angled triangle.

- (c) The quadratic equation
- $$x^2 - (2\cos\theta)x + 1 = 0$$
- has roots α and β .

(i) Find expressions for α and β in terms of θ (2)

(ii) Show that $\alpha^{10} + \beta^{10} = 2\cos(10\theta)$ (2)

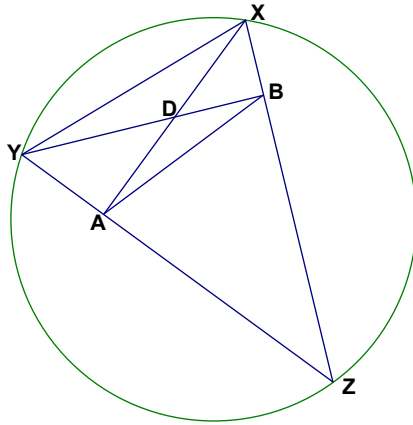
End of Question 5.

Question 6. (15 marks)

Marks

- (a) When a cubic polynomial $A(x)$ is divided by $x^2 + x + 1$ the remainder is $2x + 3$. When $A(x)$ is divided by $x(x + 3)$ the remainder is $5(x + 1)$. Find the polynomial $A(x)$ (3)

(b)



XY is a fixed chord of a circle. Z is a point on major arc XY.
The perpendicular from X to the chord YZ meets YZ at A. The
perpendicular from Y to the chord XZ meets XZ at B.
Let $\angle XZY = \theta$.

- (i) Copy the diagram into your booklet
- (ii) Prove that ABXY is a cyclic quadrilateral. (1)
- (iii) Show that $AB = XY \cos \theta$. (3)
- (iv) Deduce that as Z moves on the major arc XY the length of AB is constant. (2)
- (c) (i) Suppose that k is a double root of the polynomial equation $P(x) = 0$. Show that $P'(k) = 0$ (2)
- (ii) If $x^3 + 3px^2 + 3qx + r = 0$ has a double root prove that (4)
 $(pq - r)^2 = 4(p^2 - q)(q^2 - pr)$

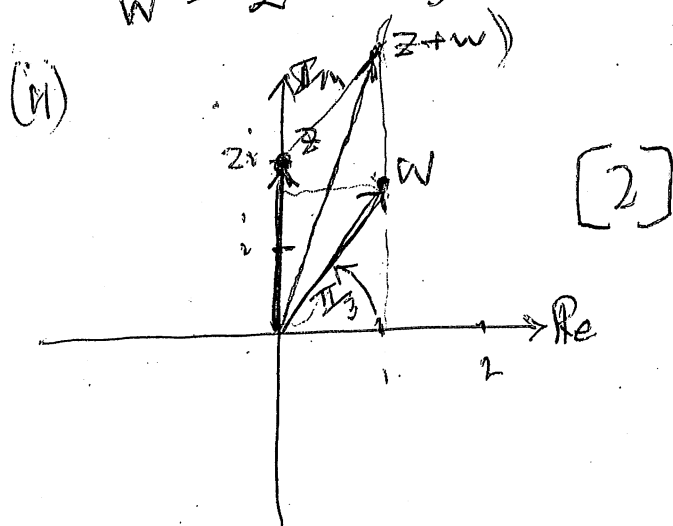
End of Question 6.**END OF SECTION C.****END OF THE EXAMINATION.**

Question 1

$$\begin{aligned}
 (a)(i) \quad iu + \bar{v} &= i(4+2i) + (3+5i) \\
 &= -2+4i + 3+5i \\
 &= 1+9i \quad [2]
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad \operatorname{Re}\left(\frac{1}{\sqrt{v}}\right) &= \operatorname{Re}\left[\frac{\bar{v}}{\sqrt{v}}\right] \\
 &= \operatorname{Re}\left[\frac{1}{|v|^2} \cdot \bar{v}\right] \\
 &= \frac{1}{|v|^2} \cdot \operatorname{Re}(\bar{v}) \\
 &= \frac{1}{34} \times 3 \\
 &= \frac{3}{34} \quad [2]
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad z &= 2i, \quad w = 1+\sqrt{3}i \\
 (i) \quad z &= 2 \operatorname{cis} \frac{\pi}{2} \\
 w &= 2 \operatorname{cis} \frac{\pi}{3} \quad [2]
 \end{aligned}$$



$$\begin{aligned}
 (iii) \quad \arg\left(\frac{z}{w}\right) &= \arg z - \arg w \\
 &= \frac{\pi}{2} - \frac{\pi}{3} \\
 &= \frac{\pi}{6} \quad [2] \\
 \arg(z+w) &= \frac{1}{2}\left(\frac{\pi}{2} + \frac{\pi}{3}\right) \\
 &= \frac{5\pi}{12} \quad [2]
 \end{aligned}$$

$$(c) \quad x^3 - 3x^2 + x - 5 = 0$$

$$\begin{aligned}
 \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} \\
 &= \frac{1}{5} \quad [3]
 \end{aligned}$$

$$\begin{aligned}
 (d) \quad \int x \tan^{-1} x \, dx \\
 &= \int \frac{d}{dx}\left(\frac{x^2}{2}\right) \cdot \tan^{-1} x \, dx \\
 &= \frac{x^2}{2} \tan^{-1} x - \int \frac{x^2}{2} \cdot \frac{d}{dx} \tan^{-1} x \, dx \\
 &= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \frac{x^2}{1+x^2} \, dx \\
 &= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \left(1 - \frac{1}{1+x^2}\right) dx \\
 &= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} x + \frac{1}{2} \tan^{-1} x + C \\
 &= \frac{1}{2} (x^2 + 1) \tan^{-1} x - \frac{1}{2} x + C \quad [3]
 \end{aligned}$$

Question 2

$$(a) \quad x^3 - 4x - 1 = 0$$

We seek eq'n w. roots $2x$ etc

$$\text{Let } y = 2x; \text{ So } x = \frac{y}{2}$$

Substitute:

$$\left(\frac{y}{2}\right)^3 - 4\left(\frac{y}{2}\right) - 1 = 0$$

$$\frac{y^3}{8} - 2y - 1 = 0$$

$$y^3 - 16y - 8 = 0$$

Replace y with x .

$$x^3 - 16x - 8 = 0 \quad [3]$$

Q2 (Contd)

(2)

(b)(i) $\sqrt{8-6i} = 3-i$ (calculator)

OR

$$\sqrt{8-6i} = a+ib$$

$$8-6i = (a+ib)^2$$

$$= a^2 - b^2 + 2iab$$

$$a^2 - b^2 = 8 \quad ; \quad 2ab = -6$$

$$ab = -3$$

$$b = -\frac{3}{a}$$

$$a^2 - \left(-\frac{3}{a}\right)^2 = 8$$

$$a^4 - 9 = 8a^2$$

$$a^4 - 8a^2 - 9 = 0$$

$$(a^2 - 9)(a^2 + 1) = 0$$

$$\therefore a = \pm 3$$

$$b = \mp 1$$

[3]

$$\sqrt{8-6i} = 3-i \text{ (Principal Root)}$$

(ii) $2z^2 + (1-3i)z - 2 = 0$

$$z = \frac{-(1-3i) \pm \sqrt{(1-3i)^2 - 4(2)(-2)}}{4}$$

$$= \frac{-1+3i \pm \sqrt{-8-6i+16}}{4}$$

$$= \frac{-1+3i \pm \sqrt{8-6i}}{4}$$

$$= \frac{-1+3i \pm (3-i)}{4}$$

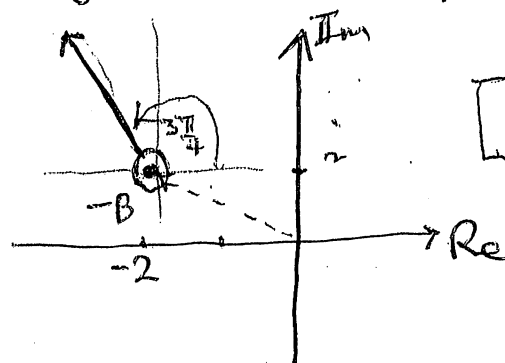
$$= \frac{1}{4}(2+2i), \frac{1}{4}(-4+4i)$$

$$= \frac{1}{2} + \frac{1}{2}i, -1+i \quad [3]$$

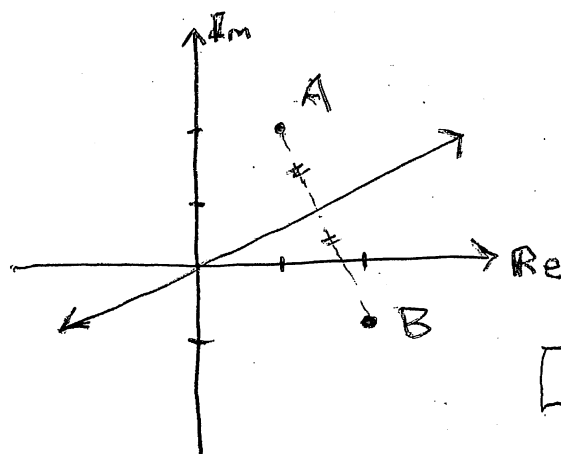
(c) $A = 1+2i$; $B = 2-i$

(i) $\arg(z+B) = \frac{3\pi}{4}$

ie $\arg(z - (-B)) = \frac{3\pi}{4}$



(ii) $|z-A| = |z-B|$



(d) $\int \frac{x+5}{\sqrt{x+6}} dx$ Let $u = x+6$
 $du = dx$

$$= \int \frac{u-1}{\sqrt{u}} du$$

$$= \int (u^{1/2} - u^{-1/2}) du$$

$$= \frac{u^{3/2}}{3/2} - \frac{u^{1/2}}{1/2} + C$$

$$= \frac{2}{3}(x+6)^{3/2} - 2(x+6)^{1/2} + C$$

$$= \frac{2}{3}\sqrt{x+6}\left(\frac{x}{3}+1\right) + C \quad [3]$$

Solutions to section (B) Ex + (II) Assessment Task (1)

Question (3) (14 mks)

$$\int \frac{dx}{\sqrt{3+2x-x^2}}$$

Now, $-(x^2-2x-3)$
 $= -[(x-1)^2-4]$
 by completing square

$$\int \frac{dx}{\sqrt{4-(x-1)^2}} \quad (2)$$

$$= \sin^{-1}\left(\frac{x-1}{2}\right) + c$$

(ii) let

$$\frac{1}{(x+1)(x^2+4)} = \frac{a}{x+1} + \frac{bx+c}{x^2+4}$$

$$1 = a(x^2+4) + (bx+c)(x+1)$$

put $x = -1$ $1 = 5a$

$$\therefore a = \frac{1}{5}$$

Equate coefficients
 of x^2 $a+b=0$
 $\Rightarrow b = -a = -\frac{1}{5}$

put $x=0 \Rightarrow c = 1 - \frac{4}{5} = \frac{1}{5}$
 $1 = 4a+c$

$$\frac{1}{(x+1)(x^2+4)} = \frac{1}{5(x+1)} - \frac{x}{5(x^2+4)} + \frac{1}{5(x^2+4)}$$

$$\int \frac{dx}{(x+1)(x^2+4)} = \frac{1}{5} \int \left(\frac{1}{x+1} - \frac{x}{x^2+4} + \frac{1}{x^2+4} \right) dx$$

$$= \frac{1}{5} \ln|x+1| - \frac{1}{10} \ln(x^2+4) + \frac{1}{10} \tan^{-1}\left(\frac{x}{2}\right) + c \quad (3)$$

(iii) $\int \frac{e^x + e^{2x}}{1+e^{2x}} dx$
 "splitting"
 $= \int \frac{e^x dx}{1+(e^x)^2} + \int \frac{e^{2x} dx}{1+e^{2x}}$
 $\downarrow$ Let $u = e^x$ $\frac{du}{dx} = e^x$
 $\int \frac{du}{1+u^2} = \tan^{-1}(e^x)$

$$\int \frac{e^{2x} dx}{1+e^{2x}}$$

Now, let $u = 1+e^{2x}$
 $du = 2e^{2x} dx$
 $= \frac{1}{2} \int \frac{du}{u} = \frac{1}{2} \ln u$
 $= \frac{1}{2} \ln(1+e^{2x})$

$$\therefore \int \frac{e^x + e^{2x}}{1+e^{2x}} dx = \tan^{-1}(e^x) + \frac{1}{2} \ln(1+e^{2x}) + c \quad (3)$$

(b) (i) $M_n = \int_0^1 (1-x^2)^{n/2} dx$
 From integration by parts:

$$M_n = \int_0^1 (1-x^2)^{n/2} \frac{d}{dx}(x) dx$$

$$= \left[x(1-x^2)^{n/2} \right]_0^1 - \int_0^1 x \cdot \frac{n}{2} (1-x^2)^{n/2-1} (-2x) dx$$

$$= n \int_0^1 x^2 (1-x^2)^{n/2-1} dx$$

$$= -n \int_0^1 (1-x^2)^{n/2-1} dx$$

3(b) (i) (cont.)

$$= -n \int_0^1 (1-x^2)^{n/2-1} dx + n \int_0^1 (1-x^2)^{n/2-1} dx$$

$$M_n = -n \int_0^1 (1-x^2)^{n/2-1} dx + n \int_0^1 (1-x^2)^{n/2-1} dx$$

$$\therefore M_n = -n M_{n-2} + n M_{n-2}$$

$$\therefore (n+1) M_n = n M_{n-2}$$

$$\therefore M_n = \left(\frac{n}{n+1} \right) M_{n-2}$$

(b) (ii) Now, $M_5 = \frac{5}{6} M_3$

$$M_3 = \frac{3}{4} M_1$$

$$M_1 = \int_0^1 \sqrt{1-x^2} dx$$

$$= \frac{\pi}{4} \text{ (from diagram)}$$

$$\therefore M_3 = \frac{3\pi}{16} \Rightarrow M_5 = \frac{5 \times 3\pi}{6 \times 16}$$

$$\therefore M_5 = \frac{15\pi}{128} = \frac{5\pi}{256}$$

Question (4) (15 mks)

(a) If $1+i$ is a root, then

$$(1+i)^3 + p(1+i)^2 + q(1+i) + 6 = 0$$

$$(1+i)^2 = 2i, (1+i)^3 = -2+2i$$

$$(q+4) + (2+2p+q)i = 0$$

$$\therefore q = -4, q+4 = 0$$

$$\text{or } 2+2p+q = 0 \quad (3)$$

$$\Rightarrow 2p = -2-q = -2+4 = 2$$

$$\therefore p = 1, q = -4$$

$$\therefore z^3 + z^2 - 4z + 6 = 0$$

is the equation. (1)

$$\sum \alpha_i = -1$$

$$(1+i) + (1-i) + \beta = -1$$

$$\therefore 2 + \beta = -1$$

$$\Rightarrow \beta = -3$$

$\therefore$ roots are $1 \pm i, -3$

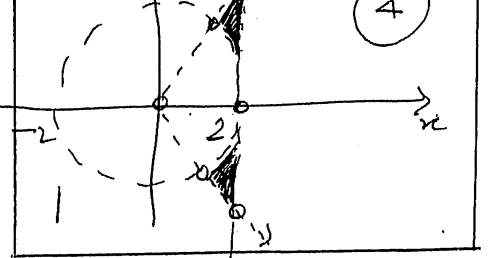
(b) $|\arg z| < \frac{\pi}{3}$

$$-\frac{\pi}{3} < \arg z < \frac{\pi}{3}$$

$$z + \bar{z} < 4 \Rightarrow 2x < 4$$

$$x < 2$$

$$|z| > \frac{2}{\sqrt{3}}$$



$x=2$

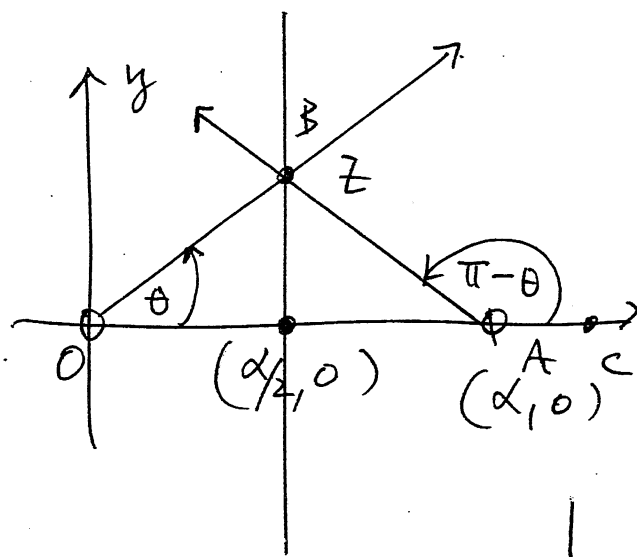
(c) $|z| = |z-\alpha|$

$\therefore$ locus is the perp. bisector joining $(0,0)$

and $(\alpha,0)$

$$\arg(z^2 - \alpha z) = \arg[z(z-\alpha)]$$

Question 4(c) (cont.)



$$\angle + \arg Z = \theta$$

$\therefore \triangle OAB$ is
Isosceles.

$$\therefore \angle BAO = \theta$$

$$\therefore \angle CAB = \pi - \theta$$

$$\begin{aligned} \therefore \arg Z + \arg(Z - \alpha) \\ &= \theta + \pi - \theta \\ &= \pi. \end{aligned}$$

$$\therefore \arg(Z^2 - \alpha Z) = \pi$$

$$(d) \quad z^5 - 1 = 0$$

4

$$\therefore z^5 = 1 = \cos(2k\pi) + i \sin(2k\pi)$$

$$\therefore z = \cos\left(\frac{2k\pi}{5}\right) + i \sin\left(\frac{2k\pi}{5}\right)$$

$$k = 0, 1, 2, 3, 4$$

$$z_1 = \cos(0) = 1$$

$$w = z_2 = \cos\frac{2\pi}{5} + i \sin\frac{2\pi}{5}$$

$$w^2 = z_3 = \cos\frac{4\pi}{5} + i \sin\frac{4\pi}{5}$$

$$w^3 = z_4 = \cos\frac{6\pi}{5} + i \sin\frac{6\pi}{5}$$

$$\begin{aligned} z &= \cos\left(2\pi - \frac{4\pi}{5}\right) \\ &= \cos\left(-\frac{4\pi}{5}\right) \\ &= \cos\frac{4\pi}{5} - i \sin\frac{4\pi}{5} \end{aligned}$$

$$\begin{aligned} w^4 = z_5 &= \cos\frac{8\pi}{5} + i \sin\frac{8\pi}{5} \\ &= \cos\left(2\pi - \frac{2\pi}{5}\right) \\ &= \cos\frac{2\pi}{5} - i \sin\frac{2\pi}{5} \end{aligned}$$

$$w^2 + w^3 = 2 \cos\frac{4\pi}{5}$$

$$w^2 \cdot w^3 = w^{2+3} = w^5 = 1$$

$$z^2 - (w^2 + w^3)z + (w^2 \cdot w^3) = 0$$

$\therefore$ real quad. equation
whose roots are w^2, w^3 is

$$z^2 - \left(2 \cos\frac{4\pi}{5}\right)z + 1 = 0$$

2009 Mathematics Extension 2 Assessment 1: **Section C** solutions

5. (a) Four medical tests A , B , C , and D are carried out within fourteen days on a patient. A must precede B and B must precede C and D . On the days when A and B are carried out the patient must not undergo any other tests, but C and D can be carried out on the same day or on different days in any order.

4

In how many ways can the days for the tests be chosen?

Solution: Method 1:

Case 1— consider when C and D are on different days. Ways of choosing the days is $\binom{14}{4}$. Then we know that the first day is A and the second is B so we have to order C and D , i.e. $2!$ ways. So the number of ways if 4 different days is $\binom{14}{4} \times 2!$.

Case 2— consider when C and D are on the same day. Ways of choosing the days is $\binom{14}{3}$.

$$\therefore \text{Total ways} = \binom{14}{4} \times 2! + \binom{14}{3},$$

$$= 2366.$$

Method 2:

A	B													
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If A and B are on the first 2 days, C can be in any of 12 places AND D can be in any of 12 places $\implies$ Ways = 12^2 .

	A	B												
--	---	---	--	--	--	--	--	--	--	--	--	--	--	--

If A and B are on the second 2 days, C can be in any of 11 places AND D can be in any of 11 places $\implies$ Ways = 11^2 ...

Hence the total number of ways with A and B together is $12^2 + 11^2 + 10^2 + \dots + 2^2 + 1 = 650$.

A		B												
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If A and B are over the first 3 days, C can be in any of 11 places AND D can be in any of 11 places $\implies$ Ways = 11^2 .

	A		B											
--	---	--	---	--	--	--	--	--	--	--	--	--	--	--

If A and B are over the second 3 days, C can be in any of 10 places AND D can be in any of 10 places $\implies$ Ways = 10^2 ...

Hence the total number of ways with A and B over three days is $11^2 + 10^2 + 9^2 + \dots + 2^2 + 1 = 506$.

$\vdots$

Therefore, the total number of ways with A and B in all possible positions is $650 + 506 + 385 + \dots + 5 + 1 = 2366$.

- (b) K is a positive real number and u, v are complex numbers. Show that the points on the Argand diagram which represent respectively the numbers:

$$u, v, \frac{u - iKv}{1 - iK}$$

form the vertices of a right-angled triangle.

Solution: Let A, B, C be the points $u, v, \frac{u - iKv}{1 - iK}$ respectively on an Argand diagram.

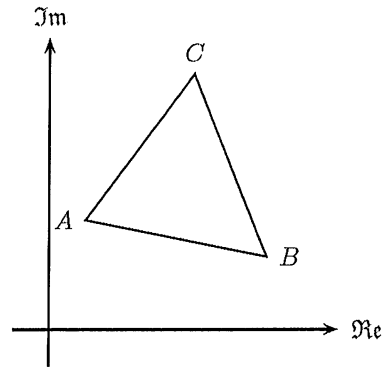
$$\overrightarrow{AB} = v - u.$$

$$\begin{aligned}\overrightarrow{BC} &= \frac{u - iKv}{1 - iK} - v, \\ &= \frac{u - iKv - v + iKv}{1 - iK}, \\ &= \frac{u - v}{1 - iK}.\end{aligned}$$

$$\begin{aligned}\overrightarrow{AC} &= \frac{u - iKv}{1 - iK} - u, \\ &= \frac{u - iKv - u + iKu}{1 - iK}, \\ &= \frac{iK(u - v)}{1 - iK}.\end{aligned}$$

$$\therefore \overrightarrow{AC} = iK \overrightarrow{BC}.$$

i.e., $\overrightarrow{AC}$ is rotated 90° from $\overrightarrow{BC}$.



- (c) The quadratic equation

$$x^2 - (2 \cos \theta)x + 1 = 0$$

has roots α and β .

- (i) Find expressions for α and β in terms of θ .

Solution: Using the quadratic equation to find x ,

$$\begin{aligned}x &= \frac{2 \cos \theta \pm \sqrt{4 \cos^2 \theta - 4}}{2}, \\ &= \cos \theta \pm \sqrt{-(1 - \cos^2 \theta)}, \\ &= \cos \theta \pm i \sqrt{\sin^2 \theta}, \\ &= \cos \theta \pm i \sin \theta.\end{aligned}$$

$$\text{So } \alpha = \cos \theta + i \sin \theta,$$

$$\text{and } \beta = \cos \theta - i \sin \theta,$$

$$= \cos(-\theta) + i \sin(-\theta).$$

- (ii) Show that $\alpha^{10} + \beta^{10} = 2 \cos(10\theta)$.

$$\text{Solution: } \alpha^{10} = (\cos \theta + i \sin \theta)^{10},$$

$$= \cos 10\theta + i \sin 10\theta \text{ by deMoivre's theorem,}$$

$$\beta^{10} = (\cos(-\theta) + i \sin(-\theta))^{10},$$

$$= \cos(-10\theta) + i \sin(-10\theta) \text{ by deMoivre's theorem,}$$

$$\therefore \alpha^{10} + \beta^{10} = \cos 10\theta + i \sin 10\theta + \cos(-10\theta) + i \sin(-10\theta),$$

$$= \cos 10\theta + i \sin 10\theta + \cos 10\theta - i \sin 10\theta,$$

$$= 2 \cos(10\theta).$$

6. (a) When a cubic polynomial $A(x)$ is divided by $x^2 + x + 1$ the remainder is $2x + 3$. When $A(x)$ is divided by $x(x + 3)$ the remainder is $5(x + 1)$. Find the polynomial $A(x)$.

3

Solution: $A(x) = (x^2 + x + 1)(ax + b) + 2x + 3,$
 $= x(x + 3)(cx + d) + 5(x + 1),$

Put $x = 0,$

$b + 3 = 5,$

$b = 2.$

Equating coefficients of $x^3, a = c.$

Equating coefficients of $x^2, a + 2 = 3c + d$ or $2a = 2 - d.$

Equating coefficients of $x, 2 + a + 2 = 3d + 5,$

$a = 3d + 1.$

Now $2a = 2 - \frac{a - 1}{3},$

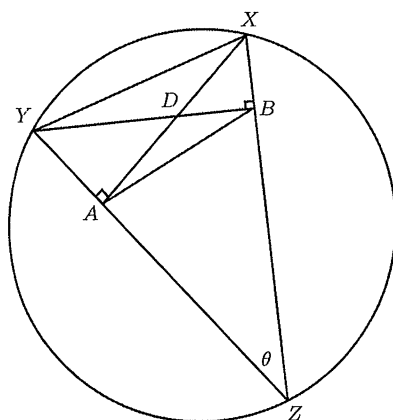
$6a = 6 - a + 1,$

$7a = 7,$

$a = c = 1.$

$\therefore A(x) = (x^2 + x + 1)(x + 2) + 2x + 3,$
 $= x^3 + x^2 + x + 2x^2 + 2x + 2 + 2x + 3,$
 $= x^3 + 3x^2 + 5x + 5.$

(b)



XY is a fixed chord of a circle. Z is a point on the major arc XY .

The perpendicular from X to the chord YZ meets YZ at A .

The perpendicular from Y to the chord XZ meets XZ at B .

Let $\angle XZY = \theta$.

- (i) Copy the diagram into your booklet.

Solution: See above.

- (ii) Prove $ABXY$ is a cyclic quadrilateral.

1

Solution: $\angle XAY = 90^\circ$ ($XA \perp YZ$),

$\angle YBX = 90^\circ$ ($YB \perp XZ$),

$\therefore \angle XAY = \angle YBX.$

$\therefore ABXY$ is a cyclic quadrilateral (angles on arc XY are equal).

(iii) Show that $AB = XY \cos \theta$.

3

Solution: $\angle ABZ = \angle XYZ$ (exterior $\angle$ theorem for cyclic quad.)
 $\angle BZA = \angle YZX$ (common),
 $\therefore \triangle ABZ \sim \triangle XYZ$ (equiangular),
 $\therefore \frac{AB}{XY} = \frac{ZA}{ZX}$ (corresp. sides of similar $\triangle$ s),
but $\frac{ZA}{ZX} = \cos \theta$,
 $\therefore \frac{AB}{XY} = \cos \theta$,
so $AB = XY \cos \theta$.

(iv) Deduce that as Z moves on the major arc XY the length of AB is constant.

2

Solution: XY is constant (given),
 θ is constant ($\angle$ s at circumf. on same arc).
 $\therefore \cos \theta$ is constant,
 $\therefore AB = \text{constant} \times \text{constant}$,
 $= \text{constant}$.

(c) (i) Suppose that k is a double root of the polynomial equation $P(x) = 0$. Show that $P'(k) = 0$.

2

Solution: Put $P(x) = (x - k)^2 Q(x) = 0$, where $Q(x)$ is a polynomial.
 $P'(x) = 2(x - k)Q(x) + (x - k)^2 Q'(x)$,
 $= (x - k)\{2Q(x) + (x - k)Q'(x)\}$.
 $\therefore P'(k) = (k - k)\{2Q(k) + (k - k)Q'(k)\}$,
 $= 0$.

(ii) If $x^3 + 3px^2 + 3qx + r = 0$ has a double root, prove that $(pq - r)^2 = 4(p^2 - q)(q^2 - pr)$.

4

Solution: Let $P(x) = x^3 + 3px^2 + 3qx + r = 0 \dots\dots\dots$ [1]
 $P'(x) = 3x^2 + 6px + 3q = 0$,
i.e., $x^2 + 2px + q = 0 \dots\dots\dots$ [2]
 $x \times$ [2] : $x^3 + 2px^2 + qx = 0 \dots\dots\dots$ [3]
[1] - [3] : $px^2 + 2qx + r = 0, \dots\dots\dots$ [4]
 $p \times$ [2] : $px^2 + 2p^2x + qp = 0, \dots\dots\dots$ [5]
[5] - [4] : $2(p^2 - q)x + qp - r = 0$,
 $x = \frac{r - pq}{2(p^2 - q)}$.
Sub. in [2] : $\frac{(r - pq)^2}{4(p^2 - q)^2} + \frac{2p(r - pq)}{2(p^2 - q)} + q = 0$,
 $\frac{(pq - r)^2}{4(p^2 - q)^2} = \frac{p(pq - r)}{p^2 - q} - q$,
 $(pq - r)^2 = 4(p^2 - q)(pq - r)p - 4q(p^2 - q)^2$,
 $= 4(p^2 - q)(p^2q - pr - p^2q + q^2)$,
 $= 4(p^2 - q)(q^2 - pr)$.